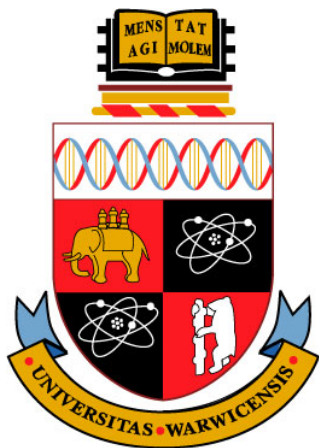




Differentiable Molecular Simulation for Solving Inverse Problems

ES98C Individual Project Proposal

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1 Introduction

Inverse problems reverse the typical forward map route to solving problems in statistical physics, inferring model parameters from empirical results. [1] With these inferred parameters, a simulator can reproduce structural observations and other measured quantities. [2] Henderson's theorem says that for a system at a fixed temperature and density, a specific pair potential $u(r)$ yields a unique pair correlation function $g(r)$. The reverse also holds true, that a measured $g(r)$ can only be produced by one unique pair potential $u(r)$. [3, 4] Despite this uniqueness, this problem is ill-posed in practice. The experimental uncertainties in values mean small perturbations have a significant influence on pair potential recovery. [5]

This project focuses on liquid structures, where two functions particularly characterise the properties of a given liquid. The radial distribution function, denoted $g(r)$, represents how particle density varies from a given reference particle. The structure factor, denoted $S(q)$, describes how a material scatters radiation, providing scientists with insights into the geometric arrangement of atoms and molecules. [6] The relationship between $S(q)$ and $g(r)$ is by a Fourier transform, however the ill-posedness of this inverse problem means that solutions are neither unique nor are they stable. [7] This project will evaluate existing work in structural scattering problems, proposing a novel, uncertainty-aware Bayesian inference approach for analysing experimental structure data.

2 Literature Review

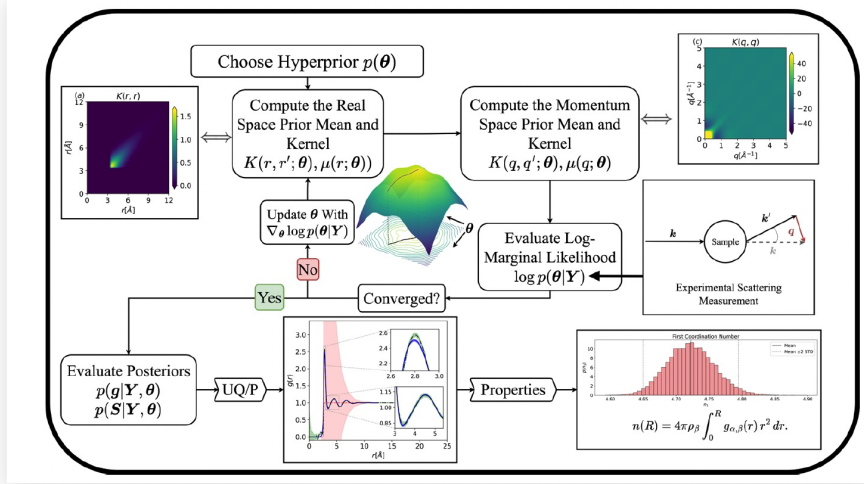


Figure 1: Flowchart visualising the “GPFT” algorithm proposed in Sullivan et al. [8]

Sullivan et al. propose a probabilistic machine learning framework that estimates partial and in some cases, full RDFs with uncertainty quantification by using nonstationary Gaussian Process (GP) regression. [8] The “Gaussian Process Fourier Transform” (GPFT) algorithm bypasses the ill-posedness of inversion from structure factor $S(q)$ to RDF $g(r)$ by using a physics-informed prior on empirical scattering data. By conditioning the mean and Gibbs kernel on empirical scattering data in real space, physical constraints such as the hard sphere repulsion ($\lim_{r \rightarrow 0} g(r) = 0$) and the long-range limit ($\lim_{r \rightarrow \infty} g(r) = 1$) are encoded directly. By exploiting the linearity of the radial Fourier transform operator $\mathcal{H}$, the Gaussian Process prior is mapped from real space to momentum space, bypassing perturbations from a general Fourier transform. Mathematically, the relationship is written as

$$S(q) = 1 + \mathcal{H}_r^{-1}[g(r) - 1], \text{ where } \mathcal{H}_r^{-1}[f] = 4\pi\rho \int_0^\infty f(r) \frac{\sin(qr)}{qr} r^2 dr \quad (2.1)$$

This linear operator means the experimental scattering data can be treated as observations from a multivariate Gaussian distribution. In turn, the posterior distribution $p(g | Y, \theta)$ can be evaluated through Bayesian inference with hyperparameter optimisation via maximising the log-marginal likelihood. Since the output is a posterior distribution rather than a point estimate, an uncertainty interval is given per prediction offering a level of confidence that is not found in deterministic approaches. Despite the predictive performance of the GPFT algorithm, it suffers from $O(N^3)$ time complexity, making it an infeasible approach for high-dimensional data. The GPFT gives the best possible reconstruction for a given liquid’s structure at fixed state (Argon at $T = 85[K]$), inferring the RDF rather than interaction parameters, meaning these predictions cannot be translated for different states. Instead of focusing on structural

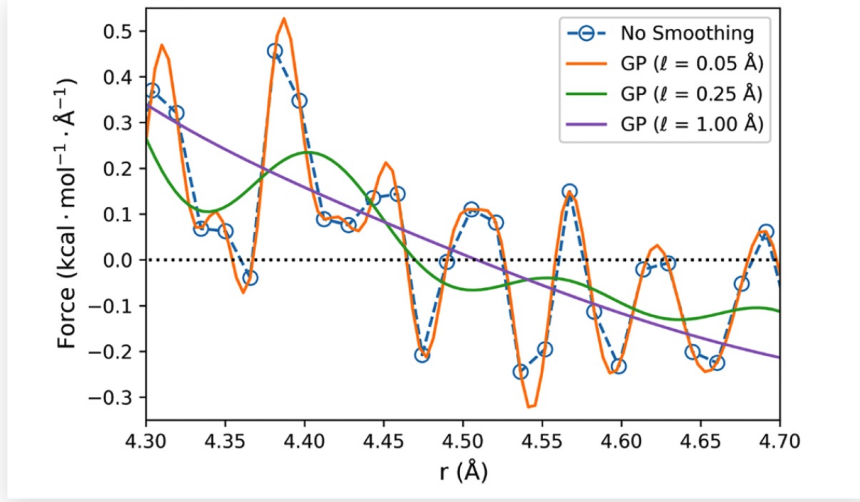


Figure 2: Force calculated from the Xe-274 K potential (blue circles) compared to GP regressed potential at length-scale parameters ($\ell = 0.05, 0.25, 1.00$ Å) [9]

inference, Shanks, Potoff and Hoefner’s “Transferable Force Fields from Experimental Scattering Data with Machine Learning Assisted Structure Refinement” focuses on the inverse problem behind deriving structure from pair potentials. [9] Their method, “Structure-Optimised Potential Refinement (SOPR),” builds on Schommer’s algorithm, an iterative approach to recover pair potentials from simulated structures [10], by using a physics-informed GP regression. The paper uses Henderson’s Uniqueness Theorem: that for pair potential $\Delta v(r)$ and pair radial distribution function $\Delta v(r)\Delta g(r) \geq 0$ holds for fixed temperature and density. This inequality is used in the linear predictor update equation

$$v_{\text{pred}}^{(n+1)}(r) = v^{(n)}(r) + \gamma k_B T [g_{\text{sim}}^{(n)}(r) - g_{\text{exp}}(r)] \quad (2.2)$$

With each update in the molecular dynamics, there is stochastic noise and finite-sampling errors that propagate through the simulation. This can result in amplifications of force calculations, reducing the robustness of the solution. [11] The paper resolves this through the use of a GP: the stochastic process outputs smooth differentiable approximations with propagated uncertainty rather than a point estimate. Smoothness is ensured by the square-exponential kernel

$$K(r_i, r_j) = \bar{\sigma}^2 \exp\left(-\frac{(r_i - r_j)^2}{2l^2}\right) + \delta_{ij}\sigma_{\text{noise}}^2, \quad (2.3)$$

where l is the length-scale hyperparameter, where l filters σ_{noise} . In addition, noise is reduced as the Gaussian form gives an analytical Fourier transformation, bypassing the perturbations when converting from real to reciprocal space.

Figure 2 highlights the noise smoothing from the GP model, showing numerical stability whilst preserving the physical behaviour of the simulation.

Nevertheless, for SOPR optimised Ne-42 K, the model predicted liquid densities with a relative error of 0.1% to 2.5%, performing as well as state of the art approaches.

The significant drawback of the SOPR approach is the use of an exhaustive forward model, with molecular dynamic simulation have a large timestep size for each iteration of optimisation. This makes scalability a bottleneck for high-dimensional parameter spaces.

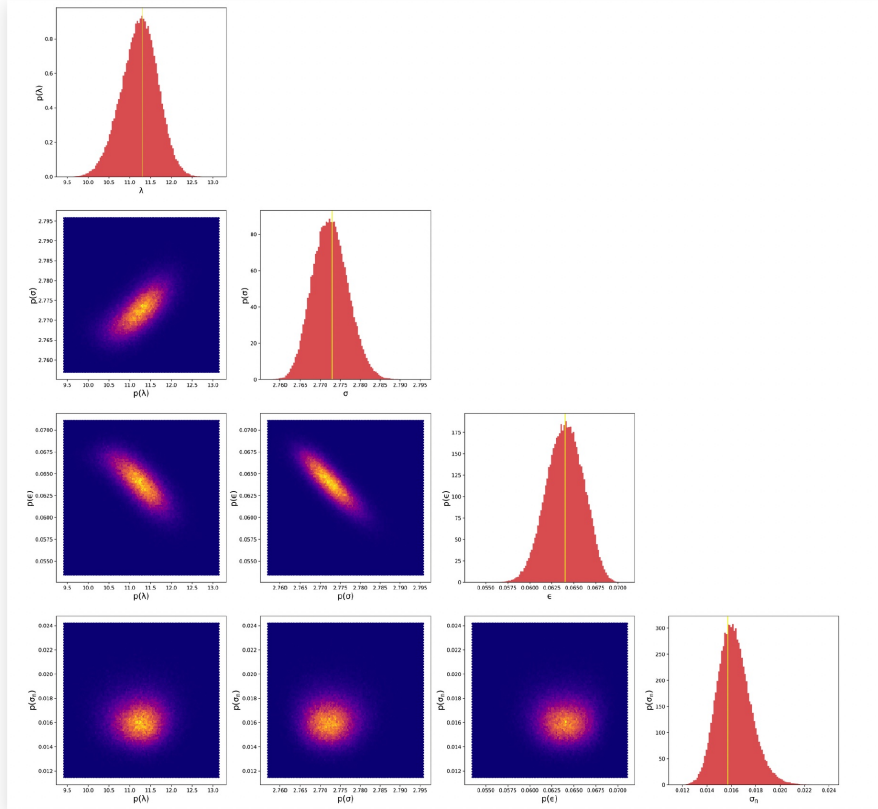


Figure 3: 1D marginal distributions for the $(\lambda - 6)$ Mie fluid parameters [12]

Shanks et al. combat the prohibitive computational cost in molecular simulation through the use of Local Gaussian Process (LGP) surrogate models. [12] The use of global GPs outputs a multivariate normal distribution for some function, i.e. the RDF, discretised into η bins, where the posterior is calculated by the equation

$$E[GP(\theta, \mathbf{r})] = \mathbf{K}_{(\theta, \mathbf{r}), \hat{\mathbf{X}}} [\mathbf{K}_{\hat{\mathbf{X}}, \hat{\mathbf{X}}} + \sigma_{\text{noise}}^2 \mathbf{I}]^{-1} \hat{\mathbf{Y}}. \quad (2.4)$$

The inversion of the kernel matrix $\mathbf{K}_{\hat{\mathbf{X}}, \hat{\mathbf{X}}}$ scales with the total number of data points (η). For N simulations, the time complexity becomes $O((N\eta)^3)$, becoming intractable for large datasets.

The LGP approach uses a conditional independence approximation: when conditioned in model parameters θ , the probability of observing a value r_i is

independent of observation r_j , forming a product of η univariate likelihoods

$$P(\mathbf{g} \mid \theta) \approx \prod_{k=1}^{\eta} P(g(r_k) \mid \theta). \quad (2.5)$$

In turn, η independent local GPs are formed, trained to map parameters to a specific radial distance rather than a vector of observations by the local expectation $E[GP_k] : \theta \mapsto S(r_k)$. More specifically, they model the difference between the simulation output y and the physics-informed prior mean μ_{prior} :

$$S(\theta, r_k) = y(\theta, r_k) - \mu_{prior}(\theta, r_k) \quad (2.6)$$

To predict the structure for the local parameters θ^* , the posterior expectation of the residual function S is conditioned on the local training inputs $\hat{X}'$ and local residuals $\hat{Y}'$, giving the equation

$$S_{local}^*(r_k \mid \theta^*) = \mathbf{K}_{\theta^*, \hat{X}'} [\mathbf{K}_{\hat{X}', \hat{X}'} + \sigma_{noise}^2 \mathbf{I}]^{-1} \hat{Y}', \quad (2.7)$$

where the local kernel matrix depends on the number of local training simulations. As a result, the global log-marginal likelihood is given by the sum of scalar log-likelihoods:

$$\log P(\mathbf{g} \mid \theta) \approx \sum_{k=1}^{\eta} \left[-\frac{1}{2} \log |\mathbf{K}_k| - \frac{1}{2} \mathbf{y}_k^T \mathbf{K}_k^{-1} \mathbf{y}_k - \frac{N}{2} \log 2\pi \right] \quad (2.8)$$

Figure 3 shows the marginal distributions calculated over the $(\lambda - 6)$ Mie potential parameters optimised to the RDF of liquid Neon. The narrow posterior distribution shows that interatomic force information is encoded within the experimental RDF. The LGP approach changes the time complexity from $O((N\eta)^3)$ to $O(N)$, with experimental speed-ups of 1.76×10^6 compared to molecular dynamics simulation, and a 2.1×10^3 speed-up compared to the global GP approach.

Whilst the computational efficiency is the primary benefit, the assumption of error independence for r_i and r_j means this low-rank approximation is susceptible to overconfident error bars. Nevertheless, there are residuals that lie beyond $2\sigma_{post}$ which the authors pin to a number of possible factors: “Fourier transform truncation, background scattering corrections, or model inadequacy.”

3 Remarks, Aims and Objectives

3.1 Remarks

The literature review demonstrates a plethora of Bayesian inference applications in materials science, ranging from nonparametric solutions to accelerated molecular dynamics simulation. The use of LGPs in particular highlights an elegant combination of Bayesian inference, molecular simulation and performance enhancement. However, the results showed a systematic error from a range of possible factors. In this ES98C Individual Project, I would like to unite the physics-informed covariance kernel proposed in Sullivan et al.’s nonparametric modelling [8] with the LGP approach, with aims to eliminate the error propagated by the Fourier transform truncation.

3.2 Aims

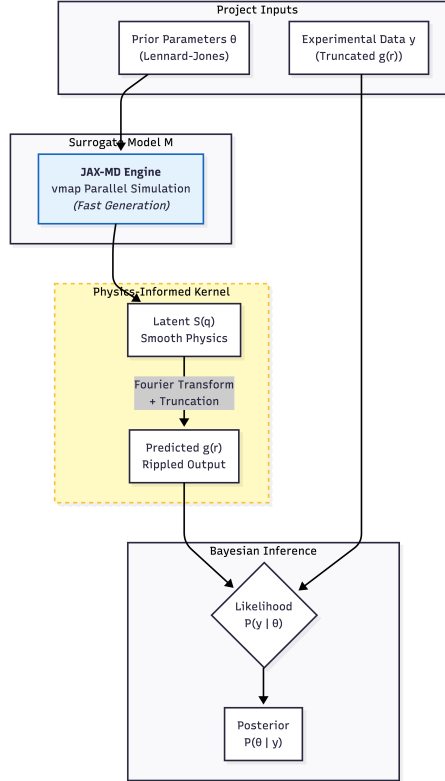


Figure 4: A high-level flowchart providing an overview for the project

This project proposes a mathematically rigorous framework for the Bayesian inference of pair potentials. The posterior distribution $P(\theta | y)$ of force field parameters θ using empirical structural data y is defined by Bayes theorem:

$$P(\theta | \mathbf{y}) \propto P(\mathbf{y} | \theta)P(\theta) \quad (3.1)$$

The likelihood $P(\mathbf{y} \mid \theta)$ is approximated via a surrogate model $\mathcal{M}$ constructed from `jax-md` simulations. [13] Whilst the LGP approach uses a stationary squared-exponential kernel for the surrogate model’s covariance, this introduces systematic error due to finite Fourier truncation.

The use of a non-stationary, physics-informed kernel as proposed in Sullivan et al’s GPFT algorithm removes this systematic error. The kernel linearly transforms the covariance of the spectral density, encoding the window function $W(q)$ into the kernel:

$$k(r, r') = \int_0^{Q_{\max}} \int_0^{Q_{\max}} \tau(q, r) k_l(q, q') \tau(q', r') dq, dq' \quad (3.2)$$

where τ is the Fourier-Bessel transform kernel (translating between reciprocal and real space), k_l is the kernel that serves as a continuous prior, representing the intrinsic smoothness of the structure factor in reciprocal space.

3.3 Objectives

The objectives of the project are as follows:

1. A molecular simulation pipeline will be created using the `jax-md` pipeline.
2. A baseline LGP approach will be used on the specified dataset, verifying the correctness of the LGP implementation via summary statistics
3. Integrate the physics-informed covariance kernel, calibrated against the simulation data, with performance verified by the same metrics as the baseline implementation.
4. Perform Bayesian inference across parameters, yielding a finalised posterior distribution with a comparison in results. Comment on whether this solves the model inadequacy of the original paper.

3.4 Expected Outcomes and Impacts

Current LGP approaches use stationary covariance kernels, showing significant performance improvements to global GP tools despite the results showing systematic error. The expected outcome of the project is a Local Gaussian Process surrogate approach that offers enhanced performance for Bayesian parameter inference whilst mitigating systematic error in Fourier-truncated data, offering a mathematically rigorous approach to modelling noise in Bayesian inference within materials science. In turn, this project hopes to offer a more robust framework for scientists to infer more accurate pair potential parameters from empirical scattering data.

4 Work Plan and Timeline

For the software segment of the project, I have drawn from different project management methods from the agile development framework. I have identified a series of deliverables (subject to change) which are delivered in two week sprints.

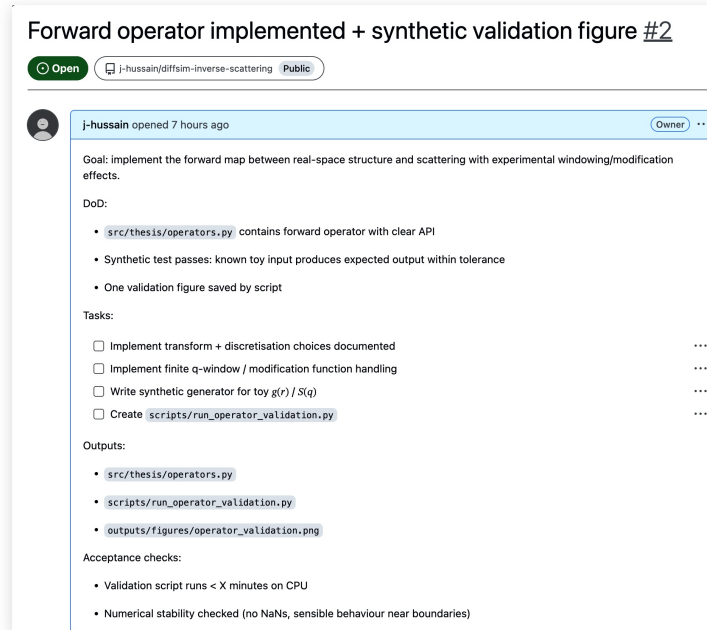


Figure 5: Example of a GitHub Issue in the project

Each deliverable is registered as an Issue on GitHub, defined by a goal, definition-of-done, tasks, expected outputs and acceptance checks. The acceptance checks are used to make sure each artefact behaves as intended, ready to move onto the next sprint of the development cycle. This approach offers iterative validation, from data preprocessing to finalised graphs.

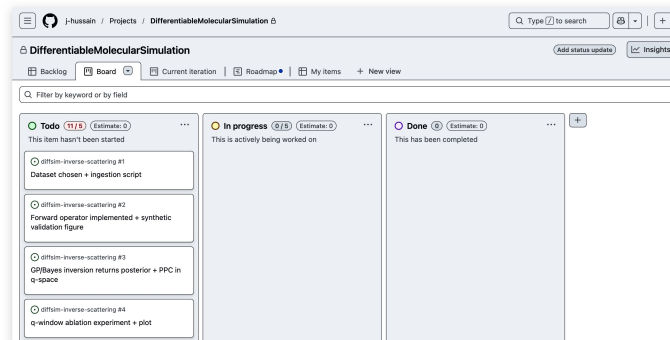


Figure 6: Example of a Kanban board within the GitHub repository

GitHub’s project management tools automatically create Kanban boards and Gantt charts to assist project management. This helps visualise the workflow for both myself and the supervisor, clearly communicating what stage of development I’m in. The timeline of the project focuses on software development and testing across June and July 2026, with August 2026 focused on report writing. I have tabulated each development sprint, listing the focus of the sprint and its corresponding acceptance check.

Sprint	Dates	Focus	Deliverables / milestone gate
Sprint 1	08–14 Jun 2026	Implement JAX-MD forward model.	Pair potential simulation (LJ/Mie/etc.) script written with <code>jax-md</code> , with data pre-processing completed.
Sprint 2	15–28 Jun 2026	Baseline LGP implementation	Simple LGP approach trained on JAX data; identify if there is systematic error in recreated experiment
Sprint 3	29 Jun–12 Jul 2026	Physics-Informed Kernel implementation.	Physics-informed covariance kernel implemented in <code>jax</code> , with a retrained surrogate. Resulting plots should showing mitigation of truncation artifacts.
Sprint 4	13–26 Jul 2026	Bayesian Inference & Evaluation.	MCMC sampler used with surrogate to yield posterior distributions $P(\theta y)$
Sprint 5	27–31 Jul 2026	Hardening & Code Freeze.	Finalised results plots with experiments comparisons. Reproducibility markdown file created.
Writing	01–31 Aug 2026	Thesis writing and final report.	Finalised thesis report.

Table 1: High-level work plan and timeline. Specific modelling choices may evolve during development; deliverables and validation gates are fixed to ensure measurable progress.

5 Budget, Constraints and Compliance

5.1 Computational Resources

- For rapid iterative development, the project will be developed on my laptop: MacBook Pro (M1 Pro), 16GB RAM.
- For larger experiments, the University of Warwick’s Scientific Computing RTP (SCRTP) offers high performance computing (HPC) facilities. The SCRTP’s *Avon* HPC cluster, comprised of 16 nodes, with a large CPU pool and 3 NVIDIA RTX 6000 GPUs per node subject to allocation. With this cluster, JAX-MD simulations in parallel, with significantly faster hyperparameter tuning as opposed to running it on a local CPU/GPU.

5.2 Software Resources

- The programming language of choice is Python. Python is open-source, so there are no licensing costs.
- The Python libraries that will be used are from the JAX ecosystem (`jax`, `jax-md`, `equinox`) or general open-source data science libraries (`pandas`, `matplotlib`). There are no associated costs.
- GitHub will be used for version control and traceability.

5.3 Data Resources

- Open datasets will be used, as paid data access is outside the scope of the budget. **Mention specific dataset – provide link**
- Should data redistribution be restricted by file size or access, the GitHub repository will host accompanying documentation to retrieve the dataset.

5.4 Ethical Considerations

This project has no use of human data, nor does it have applications relating to human science (medicine, social sciences, etc.) and thus an ethical assessment is not required.

5.5 Risk Assessment

Hazard	Risk Level	Mitigation Strategy
Physical Limitations	Medium	Long coding sessions can cause eye strain, repetitive strain injury (RSI), posture pain. Abiding by the 20-20-20 rule should mitigate eye strain, and ergonomic desk equipment will be used where possible.
Hardware Overheating	Low	Intensive GPU usage for training may cause hardware heating. Try to gain access to the SCRTP Avon cluster, which is professionally cooled. Otherwise, use batch computing to mitigate overheating.
Data Loss	Medium	Drive failure or corruption may cause significant setbacks. Overleaf will be used for thesis writing, with local copies of document files in case of Overleaf services being terminated. Use GitHub for documenting all other files.

Table 2: Risk assessment for the thesis.

References

- [1] Richard Kwame Ansah, Kassim Tawiah, and Ruth Naayi Oduankey Abbey. Advancing the inverse problem in statistical mechanics: A five-body interaction perspective. *Physica A: Statistical Mechanics and its Applications*, 674:130700, 2025.
- [2] Marc C. Kennedy and Anthony O’Hagan. Bayesian calibration of computer models, 2001.
- [3] R.L. Henderson. A uniqueness theorem for fluid pair correlation functions. *Physics Letters A*, 49:197–198, 1974.
- [4] Markus Deserno. The henderson theorem, August 2018.
- [5] Haina Wang, Frank H. Stillinger, and Salvatore Torquato. Sensitivity of pair statistics on pair potentials in many-body systems. *The Journal of Chemical Physics*, 153, 2020.
- [6] P. Chieux. Introduction to accurate structure factor measurements of disordered materials by neutron scattering. *Journal of Molecular Structure*, 296:177–198, 1993.
- [7] V. Sánchez-Gil, E.G. Noya, L. Temleitner, and L. Pusztai. Reverse monte carlo modeling: The two distinct routes of calculating the experimental structure factor. *Journal of Molecular Liquids*, 207:211–215, 2015.
- [8] Harry Winston Sullivan, Matej Cervenka, Brennon L. Shanks, and Michael P. Hoepfner. Physics-informed gaussian process inference of liquid structure from scattering data. *The Journal of Physical Chemistry B*, 129:11802–11815, 2025.
- [9] Brennon L. Shanks, Jeffrey J. Potoff, and Michael P. Hoepfner. Transferable force fields from experimental scattering data with machine learning assisted structure refinement. *The Journal of Physical Chemistry Letters*, 13:11512–11520, 2022.
- [10] W. Schommers. Pair potentials in disordered many-particle systems: A study for liquid gallium. *Physical Review A*, 28:3599–3605, 1983.
- [11] D. Levesque, J. J. Weis, and L. Reatto. Pair interaction from structural data for dense classical liquids. *Physical Review Letters*, 54:451–454, 1985.
- [12] Brennon L. Shanks, Harry W. Sullivan, Abdur R. Shazed, and Michael P. Hoepfner. Accelerated bayesian inference for molecular simulations using local gaussian process surrogate models. *Journal of Chemical Theory and Computation*, 20:3798–3808, 2024.
- [13] Samuel S. Schoenholz and Ekin D. Cubuk. arxiv query: `searchquery = amp; idlist = 1912.04232amp; start = 0amp; maxresults = 10`, 2019.